Adeesh Desai

Founding Research Engineer & Technical Lead, Factory Intelligence | M.S. Robotics, Purdue University

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RESEARCH INTERESTS

Multimodal models for loco-manipulation, tactile sensing, and reinforcement-learning fine-tuning of robot foundation models. My work asks how robots can use, or imagine, the sense of touch for contact-rich manipulation, and what deploys reliably in factories.

EDUCATION

Purdue University, West Lafayette, IN Aug 2024 – May 2026
M.S.E. in Robotics · Graduate Research Assistant, MARS Lab (advisor: Prof. Yu She)

M S Ramaiah Institute of Technology, Bengaluru, India Aug 2019 – May 2023
B.E. in Mechanical Engineering

EXPERIENCE

Factory Intelligence: Founding Research Engineer & Technical Lead Aug 2026 – Present · Lafayette, IN

- First research hire at an early-stage startup automating factory tasks with robot learning; technical lead for the robot-learning effort, leading a small team from data collection through policy training to on-site deployment.
- Built real-time teleoperation and data-collection pipelines for bimanual arms (VR and leader–follower), with synchronized multi-camera and proprioceptive logging.
- Fine-tune vision-language-action and diffusion-based manipulation policies on in-house demonstrations, and reinforcement-learning fine-tuning of pretrained policies for contact-rich assembly; deployed learned policies on real robot cells and in production factories.

MARS Lab, Purdue University: Graduate Research Assistant Jan 2025 – May 2026 · West Lafayette, IN

- **TacImag** (co-first author): tactile imagination framework that predicts tactile observations from vision and proprioception with a conditional diffusion model, enabling touch-informed manipulation with no tactile sensor at deployment (+44.4 / +23.3 pts on real-robot contact- / texture-sensitive tasks).
- **CONTACT** (co-first author, IROS 2026): benchmark of ten simulated and real disassembly tasks showing when tactile sensing matters; structured force-field representations consistently outperform vision-only and tactile-image baselines.
- **ContactWorld**: vision–tactile world-model benchmark across 12 contact-rich tasks; point-cloud + tactile force-field representations give the strongest long-horizon planning.
- Large-scale simulation studies and real-robot experiments (Franka, UR5e) with GelSight-style tactile sensors; built tactile simulation and policy-learning infrastructure.

e-Yantra, Embedded Systems & Robotics Lab, IIT Bombay: Project Engineer May 2023 – Jul 2024 · Mumbai, India

- Designed modular multirotor drones (3.7″–10.6″ wheelbases) for the “Drone in a Box” research platform; integrated GPS, magnetometer, and optical-flow LiDAR (I2C/SPI) for outdoor navigation and return-to-home; tuned Betaflight/ArduPilot firmware; mentored interns on Kalman-filter and PID control.

PUBLICATIONS

Imagining the Sense of Touch: Touch-Informed Manipulation via Imagined Tactile Representations. Z. Zhang*****, **A. Desai***, J.-C. Hu, Y. Saka, Q. K. Luu, J. Lei, D. Soleymanzadeh, B. Zhang, M. Zheng, Y. She. Under review, 2026 · IROS 2026 Workshop on Scalable Tactile Sensing for Dexterous Manipulation · arXiv:2607.01684

CONTACT: CONtact-aware TACTile Learning for Robotic Disassembly. Y. Saka*****, J.-C. Hu*****, **A. Desai***, Z. Zhang*****, B. Zhang, Q. K. Luu, M. R. I. Prince, M. Zheng, Y. She. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026 · arXiv:2603.08560

ContactWorld: What Matters in Vision-Tactile World Models for Contact-Rich Manipulation. Z. Zhang*****, P. Zhou*****, K. Zhang, **A. Desai**, T. Amosa, D. Soleymanzadeh, J. Lei, M. Zheng, Y. She. Under review, 2026 · IROS 2026 Workshop on Scalable Tactile Sensing for Dexterous Manipulation · arXiv:2606.13877

***** *equal contribution*

AWARDS & SERVICE

- **3rd Place, “What Bimanual Can Do” (WBCD) Competition, ICRA 2025.** Logistics Packing track, $5,000 prize; VR teleoperation framework (Meta Quest 3 + Franka) with an operational-space controller.
- **Reviewer:** IEEE Robotics and Automation Letters (RA-L), 2026– · IEEE ICRA 2027.

SKILLS

Robot learning	Imitation learning, diffusion policies, VLA fine-tuning, RL / HIL-RL fine-tuning, world models, tactile representation learning
Systems	ROS/ROS2, real-time teleoperation (VR, leader–follower), multi-sensor data collection, CAN-bus actuators, real-robot deployment (Franka, UR5e, bimanual arms)
Software	Python, C/C++, PyTorch, Isaac Gym, MuJoCo, OpenCV, Git, Linux
Hardware	Tactile sensors (GelSight-style, piezoresistive), drone design, embedded sensing (I2C/SPI), SolidWorks/CAD